

7.3 The management of urban growth

This chapter will explain:

- ★ the strategies and techniques used to manage urban growth
- ★ the extent to which these strategies and techniques are successful
- ★ how the management of urban areas can be sustainable
- ★ detailed specific example: urban change in Shanghai.

Several issues in urban areas need to be managed, including housing, transport, industry, retail, leisure, social issues such as migration and inequality, and environmental issues such as air pollution and access to clean water. Many strategies and techniques can be used.

Managing housing

Access to affordable, high-quality housing is one of the biggest issues in urban areas. This is due in part to in-migration, but also to more people living longer and households splitting up. Potential solutions include building more social housing that is affordable for more people. However, developers make greater profits from building more expensive housing for richer residents. To deal with more single people in urban areas, more flats and apartments could be built.

Renewal of older buildings (gentrification) allows some buildings to be re-used rather than demolished. However, it may force poorer residents out of a neighbourhood. In some areas, old industrial buildings can be converted into residential buildings. For example, Lucy's Iron Foundry in Oxford was converted to residential flats ([Figure 7.16](#)). This can be a sustainable use of an old, derelict building.

In many LICs, squatter settlements (low-income housing) could be given legal status/security of tenure, which would remove the threat of them being demolished by the local or national government. In many HICs, such as the USA and the UK, dealing with the increased demand for housing in large urban areas has led to the decentralisation of people away from larger urban areas to smaller urban areas, including new towns and expanded towns.



▲ **Figure 7.16** The conversion of Lucy's Iron Foundry from an industrial to a residential use

Managing transport

Managing transport can be a major issue in many urban areas. Transport can lead to congestion, poor air quality and, in some cases, road rage. However, there are many ways to reduce the number of vehicles on the roads in urban areas and many examples of sustainable forms of transport management. These include:

- » encouraging the use of public transport
- » the creation of low-emission zones
- » car-pooling schemes (such as in BedZED in London)
- » park-and-ride schemes (such as in Oxford, UK)
- » congestion charge schemes
- » vehicle exclusion zones
- » use of vehicle-restriction practices (e.g. only allowing vehicles with an odd number plate on certain days of the week and with an even number plate on the other days, as used in Mexico City, Athens and Singapore)
- » the reintroduction of trams (e.g. between Wimbledon and Croydon in London)
- » permit-only parking
- » pedestrianised shopping streets (such as Rose Street, Edinburgh)
- » cycle lanes (such as in Copenhagen)
- » transport calming (e.g. speed bumps)
- » speed limits.

Singapore opened a Mass Rapid Transit (MRT) system in 1987. This has proved very successful and has enabled population and industry to move away from central areas but still be able to reach the city centre quickly. There are six MRT lines, over 130 stations and about 200 km of rail line. The MRT system and the bus network meet at Integrated Transport Hubs. The government has also tried to reduce the number of private vehicles on the road by introducing a system that allows different vehicles to use the roads only on certain days of the week.

Managing industry

Large-scale industrial polluters can be moved, for example the steel works in Beijing was relocated away from the city prior to the 2008 Olympic Games. In many HICs there has been a large-scale decline of industry in urban areas, notably the car industry in Detroit. The change from manufacturing industry to service industries tends to have a positive impact on the natural environment. Many science parks are landscaped, with green spaces and water features.

A more recent development, which accelerated during the Covid-19 pandemic, was the number of people working from home. This trend was more noticeable in HICs than LICs due to the greater coverage of, and access to, ICT. It reduces the number of people commuting to work and, arguably, is more sustainable.

Managing retail

Issues relating to retailing include the higher cost of living in many large urban areas, the development of more out-of-town retailing, which benefits car owners, and the decline of many more central retail units due to the developments on the urban fringe. This relates mainly to HICs that have a large proportion of car-owning households. Many retailing premises have closed in recent years, in part due to the growth of online retailing and in part due to the Covid-19 pandemic. Retailing in cities is very diverse. At one end, there are formal shopping areas, such as in city centres and retail parks. At the other end, much retailing is informal and carried out by individuals who sell goods by the roadside. Some markets may be permanent, whereas others might occur at a location once a week or once a month. Some 'retailing' may occur at car-boot sales, where individuals or households sell some of their own goods to other people.

Managing leisure

In some urban areas, the amount of open space available for recreation and sport is limited. This is true in LICs and in some HICs. Many schools have limited areas for sports to take place. However, in areas of low-density urban development, such as in the USA and Australia, there is more open space available for leisure facilities. Some forms of leisure can be very large-scale, such as national sports events or professional sports teams, and these may generate large numbers of supporters to watch the sporting events. In many cases, this may lead to congestion, litter, and noise and light pollution. However, such events may provide great enjoyment, much employment and social/community activities for local residents.

Many leisure activities are very sustainable and have limited environmental impact, for example walking, jogging and open-water swimming. Some forms of leisure may be very productive, such as the provision of allotments so that people can produce food. Some food may also be grown in people's gardens, for example by having fruit trees.

Managing social issues

Population issues in large urban areas include exclusionary zoning, in which low-income people of selected racial groups are prevented from living in certain areas. This is also called 'red-lining'. Urban areas may be

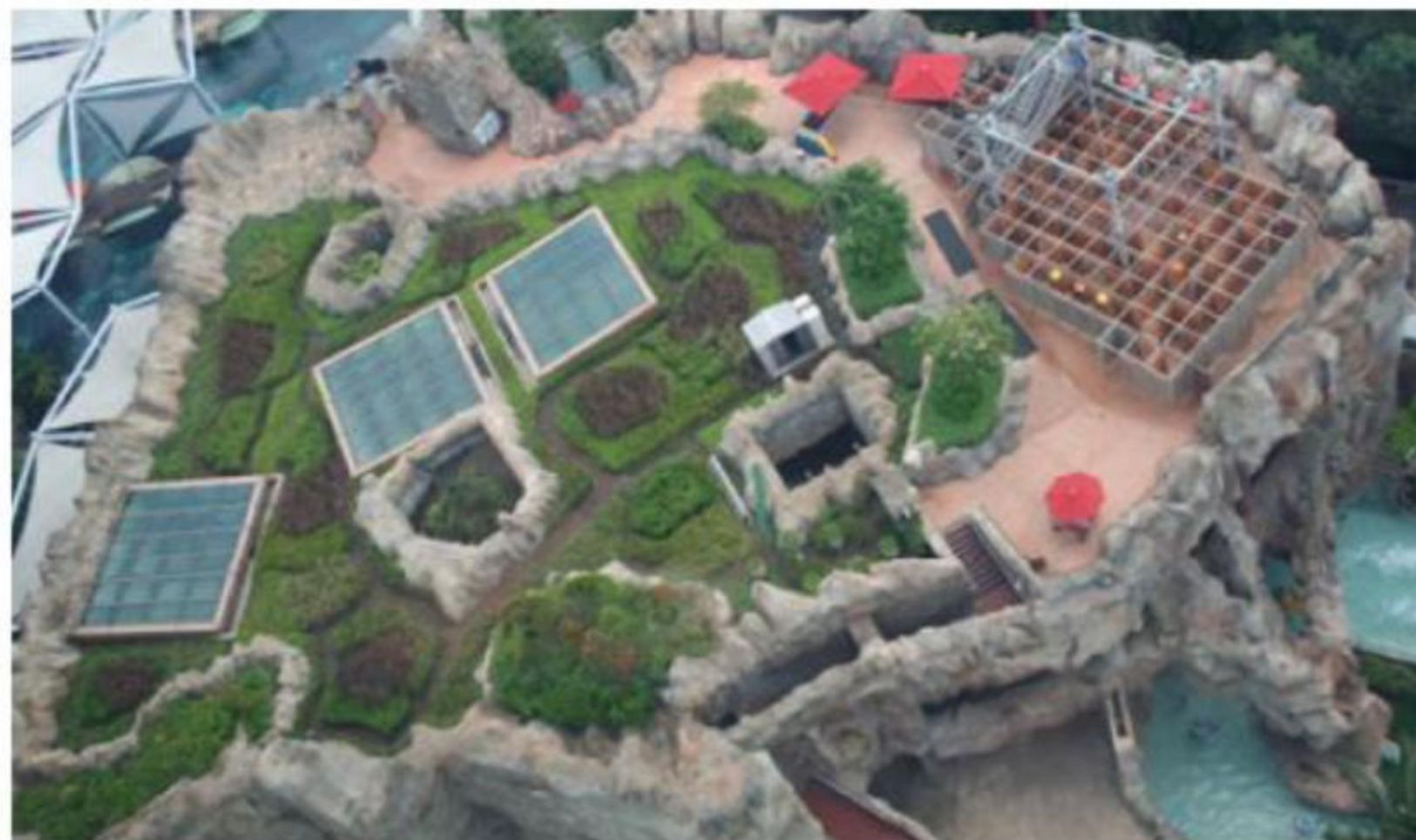
considered as ‘melting pots’ for migrants or may lead to anti-migrant feelings. Urban poverty is another urban issue. This could be addressed by creating more jobs and building partnerships between higher and lower-income areas, for example by working together to improve informal settlements or expanding housing for those in poverty. However, for many urban areas this is too costly or not a priority.

Managing environmental issues

Air pollution in urban areas can be an issue due to the concentration of industries, factories and vehicles. In addition, some households burn fuelwood, rubbish and plastic indoors, or use paraffin or low-grade coal. Potential solutions include planting more trees, having fewer vehicles on the road and using renewable energy sources rather than coal or fuelwood.

In Warsaw, Poland, the burning of coal was banned in 2023 in order to reduce air pollution. Pollution controls need to be established and enforced. In India, Prime Minister Modi has pledged to reduce the country’s carbon emissions by 45 per cent by 2030. However, coal accounts for about 50 per cent of energy in India. In urban areas, such as Mumbai and New Delhi, authorities have taken to washing some roads and building sites to improve air quality. New Delhi is also affected by the burning of wheat and rice in winter. India’s National Clean Air Programme was introduced in 2019. Initiatives include promoting renewable energy, developing electric vehicles and encouraging households to use liquefied petroleum gas for cooking.

Water in many urban areas may be scarce due to the large number of people present and/or climate change, as was seen in Cape Town’s water shortage (‘Day Zero’), which peaked in 2017–18. To overcome this, more water could be collected in water butts, greywater could be used for watering gardens and/or impermeable surfaces could be changed to permeable ones so that water does not run off. Programmes to increase the amount of urban forests or green roofs ([Figure 7.17](#)) could store more water in urban areas, and there could be more water conservation in homes, schools and industries.



▲ **Figure 7.17** Green roofs in Singapore

To improve urban environments, the number of vehicles could be restricted. New open spaces could be created, such as the Queen Elizabeth Park in East London for the London 2012 Olympic Games. Other ways of creating open spaces include green belts, green wedges and green spaces.

Activities

- 1 Evaluate two strategies used to manage urban growth.
 - 2 Compare the strategies to manage either housing or transport in urban areas in LICs and HICs.
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Detailed specific example

Urban change in Shanghai

Urban growth in Shanghai

For around 700 years Shanghai has been one of Asia’s major ports, and it has had a varied history. It thrived until 1949, when China closed itself off from trade with the West. This changed Shanghai from an international centre of production and trade to an inward-looking city. During the 1970s, China began slowly reopening its economy to the world, and Shanghai was designated one of 14 open cities. The Shanghai Economic Zone was established in 1983, and in the early 1990s an ambitious major programme of redevelopment was started, especially in the eastern hinterland around Pudong ([Figure 7.18](#)). The neighbouring cities of Suzhou and Wuxi have slowly merged with Shanghai to create one continuous megalopolis. Since economic reforms began in China in 1978, around 200 million people have moved from rural areas to urban areas. This may be the largest population movement in human history.

Site and situation

Shanghai developed on a flat, low-lying alluvial plain on the banks of the Yangtze River ([Figure 7.19](#)). It is located at the confluence of the Huangpu and Suzhou Rivers – so it has an excellent location for shipping and trade. From 1844, British, French, American and Japanese traders owned land in Shanghai. By 1920 it was China’s largest and most important city,

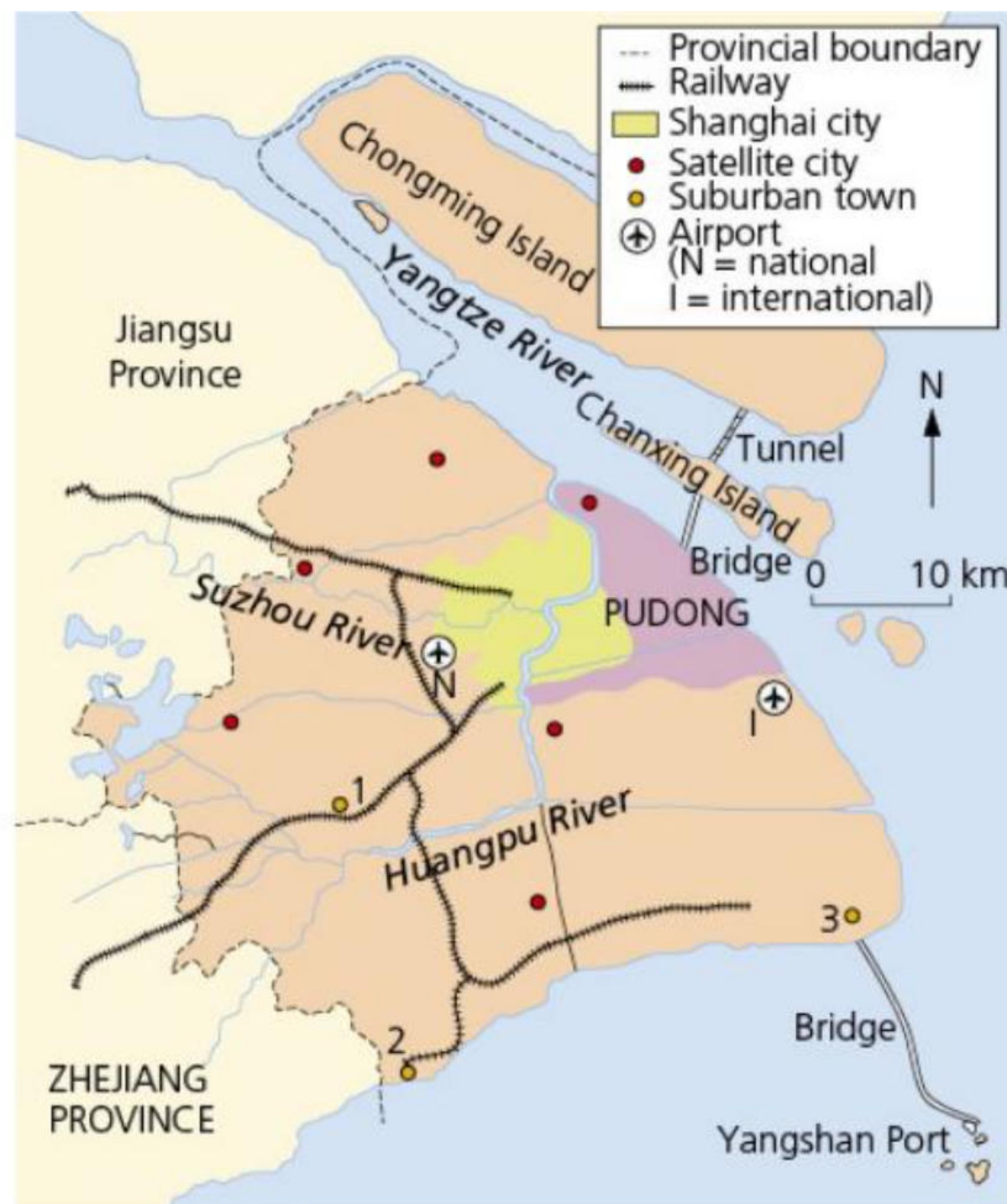
but the foreign influence declined after the end of the Second World War and following the Communist Revolution in 1949.



▲ **Figure 7.18** Housing in Pudong, Shanghai



▲ **Figure 7.19** The situation of Shanghai in China



▲ **Figure 7.20** Land use in Shanghai

Economic change

Between 1949 and 1976, political influences such as the Great Leap Forward (1958–60) and the Cultural Revolution (1966–76) focused attention away from rural areas, foreign influence and capitalist development. During this period 1 million people returned to the countryside.

However, in 1979 the first generation of Special Economic Zones was created. Although Shanghai was not one of them, it benefited from relaxed housing restrictions such as the subdivision and subletting of housing.

In 1984, Shanghai was declared open to foreign development. For much of the next 20 years, Shanghai's economic growth rate was over 12 per cent per annum. In 1990, a new CBD was created in the Pudong area ([Figure 7.20](#)). Banks, stock exchanges and insurance companies moved in. By 2000, over 3000 skyscrapers had been built, including the Shanghai Financial Centre.

Having established a strong industrial base, by 1990 the city was well placed to take advantage of the new opportunities offered by globalisation. It became a major centre for export manufacturing (based on automobiles, biotechnology, chemicals and steel), and its service industry sector (trade, finance, real estate, tourism and e-commerce) helped to diversify its economy. Between 1990 and 2000, Shanghai began to re-emerge as a world city. Foreign investment was attracted. Over half of the world's top 500 transnational companies and 57 of the largest industrial enterprises set up in Shanghai, contributing to an annual regional growth rate of over 20 per cent – more than twice the national average. In 2009, Shanghai was ranked the seventh-largest city in the world with a population of 15 million. In 2024 it had grown to almost 25 million.

Since 1990, the city's manufacturing sector has steadily contracted, losing over 1 million jobs, while the business services, finance and real estate sectors have expanded. Rising demand for highly skilled labour has led to further in-migration, resulting in an increasing disparity in wealth. Shanghai's experience does lend support to the general hypothesis that world city status inevitably leads to a widening gap between lower and higher incomes.

In 2006, Yangshan Port was opened to accommodate larger ships that could enter the Huangpu and Suzhou Rivers. Yangshan was built on an island about 40 km southeast of Shanghai and connected by a 35 km bridge. It is now the world's largest port. In 2023, it handled over 49 million containers, compared with 39 million in Singapore, the world's second-largest port.

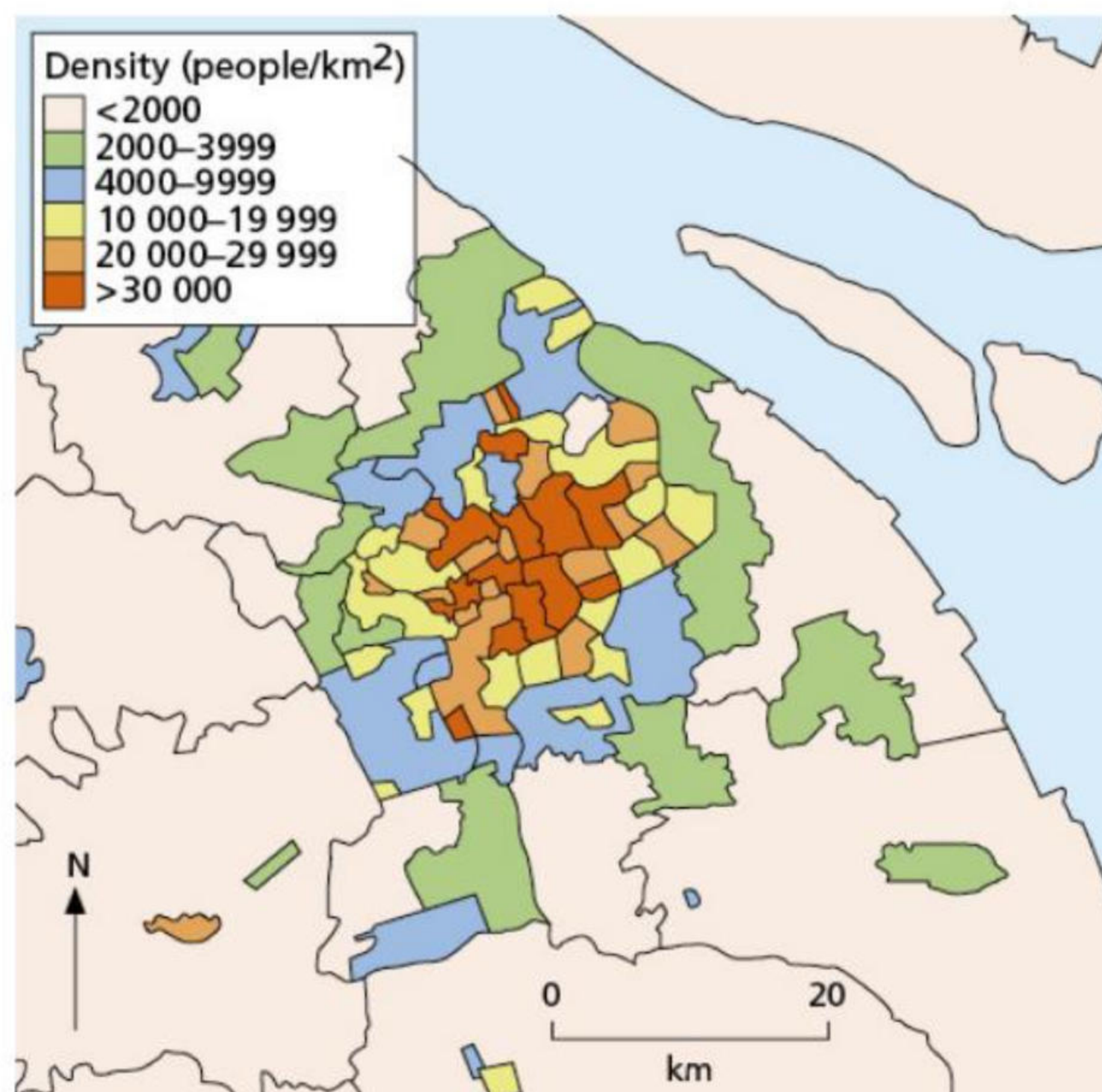
Shanghai is a city-state. It is part of the Yangtze River Delta Economic Zone, one of the fastest-growing urban areas in the world, containing 16 megacities, including Shanghai. The region has a population of 75 million and earns 25 per cent of China's GNP – and 50 per cent of its foreign direct investment. The city has been described as the largest construction site in the world – in 2020, 4000 buildings with over 24 storeys were under construction.

The challenges and opportunities of urban growth - and the strategies used to manage it

Housing and demographic issues

Housing shortages and overcrowding problems are acute. Almost half of the population live in less than 5 per cent of the total land area, and in central Shanghai population density reaches 40,000–160,000 people per square kilometre ([Figure 7.21](#)). Population pressure is caused by in-migration, overcrowding, disparities in wealth and the social insecurity of Shanghai's 'floating population'. From the 1990s, whole neighbourhoods were demolished. Over 2 million residents were moved to the outer suburbs to live in better-quality accommodation. Many people are unaware of their property rights. As property prices increase, they are given insufficient compensation and they cannot afford alternative housing in their

old neighbourhoods.



▲ **Figure 7.21** Population density in Shanghai

The Shanghai government has established a series of important policies to address these problems:

- A combination of widespread family planning and medical care, which has controlled fertility levels among the young immigrant population
- Compulsory work permits
- Educational initiatives to improve immigrant job opportunities.

These programmes have reduced population density in the heart of the city and increased it in the suburban satellite cities such as Songjiang. This has been a successful strategy, although it means that many who have been moved out under the decentralisation policy must travel back to the centre for work.

Economic growth has attracted an increasing number of Chinese living overseas and foreign migrants to live in Shanghai. Many of these live in luxurious, gated apartments.

Changes in transport infrastructure in Shanghai

As a result of population growth and the expansion of the built-up area of Shanghai, there is the need to develop transport systems. Shanghai's integrated transport system is developing a transport infrastructure that will deal with increased traffic volumes. The integrated transport system focuses on two ports, two highways and three transport networks.

Shanghai has completed the north port area of the Yangshan Port. It is the busiest port in the world, with an annual cargo of nearly 700 million tonnes. Shanghai has two international airports and four airport terminals. It receives over 70 million passengers each year. Pudong International Airport is the world's third largest in terms of cargo.

Shanghai has over 12,000 km of roads, including nearly 800 km of expressways. The target for planners has been referred to as '15, 30 and 60', meaning that motorists in suburban areas can reach an expressway within 15 minutes, then travel to the city centre in 30 minutes, and the time needed to travel between any two suburban areas is less than 60 minutes.

The Urban Expressway Network is over 4000 km long, with over 3000 km of road in the city centre and nearly 200 km of expressway. There are some 23 crossing facilities for the Huangpu River, including four bridges and 12 tunnels.

Shanghai's railway has nearly 400 km of line. Journey times have been shortened – for example, journey time fell from 90 minutes to 30 minutes to Hangzhou and from 180 to just over 70 minutes to Nanjing. The urban rail network has developed in less than 20 years. There are 13 Metro lines, which together carry over 5 million passengers daily. Approximately 25 per cent of the city centre is covered by railway stations, serving 40 per cent of the city centre's population. In addition, there are over 1000 bus lines and 17,000 buses.

The practical targets of Shanghai's transport strategy are that:

- public transport between any two parts of the city centre should take less than 60 minutes
- residents in suburban areas should be able to access the Metro with just one bus ride
- there should be direct bus links between the new towns and the urban centre.

Access to water and sanitation

The Huangpu River and the Yangtze River are the main surface water sources for Shanghai's water supply. Parts of the upper reaches of the Huangpu River pass through suburban Shanghai, with its intensive farming activities, while the lower course passes through urban areas, with intensive industrial activities.

In urban areas throughout China, the proportion of people with access to piped water rose from 40 per cent in 1990 to

nearly 99.4 per cent in 2022. By 2020 nearly 90 per cent of Shanghai's wastewater was treated. Over 70 per cent of households have access to sewerage services.

Although water is abundant in Shanghai, there are high levels of water stress. Increasing demand, pollution and saltwater intrusion are threatening the city's water security. For example, agricultural practices have led to fertilisers and insecticides getting into the urban water system. The relatively low capacity of the sewage treatment system has led to some industrial and residential waste being discharged directly into rivers. Due to organic pollution, the Huangpu River has lower water quality than the Yangtze River. In addition, saltwater intrusion is a seasonal problem in the winter and early spring dry season.

To combat the need for more freshwater, Shanghai built the Qingcaosha Reservoir, and this has been designed to provide water for up to 68 days (the theoretical maximum extent of saltwater intrusion in Shanghai).

Waste treatment

As Shanghai increases in size, it is producing more rubbish, and the sites for landfills are filling up. Increasingly, Shanghai is turning to incineration, and generating electricity at 'waste to energy' plants, such as the one at Hangzhou. The demand for such incinerators is increasing at a much more rapid pace than elsewhere in the world.

In the past, most of Shanghai's rubbish ended up in landfill sites or in unregulated heaps on the edge of the city. At such sites, there was much informal recycling of materials, but there was also contamination of the land and groundwater by methane. Plastics were carried into the rivers, and from there into the ocean.

Shanghai currently produces the most household rubbish in China – 22,000 tonnes/day. China's largest landfill, Laogang, is located near the city by the coast. However, there is now an incinerator next to the landfill and this takes in around 3000 tonnes a day. This is much more sustainable. The waste is burned at high temperatures (850°C or higher) to destroy toxins. Water is heated to produce steam, which in turn drives the turbines to generate electricity. The new incinerators at Laogang will burn 9000 tonnes a day. The aim is to increase the amount of waste incinerated from 35 per cent of domestic waste to 75 per cent.

Shanghai's Master Plan

According to Xie and Zhang (2020), the Shanghai Master Plan (introduced in 2000) focused too much on westernisation and economic growth (for which it has been very successful) and not enough on human development. They consider that Shanghai's main aim in the 2017–35 Master Plan is to be an 'excellent global city', but that growth and development have led to increased social disparities.

The UN had previously described Shanghai as the model city for sustainable development, with its 2011 'Better city, better life' programme promising balance, growth and an open and multicultural society. However, developments have been dominated by Western-dominated practices, notably a concentration on economic development. This has reinforced gaps in social and environmental justice. China's income inequality is high – the top 10 per cent of households account for 30 per cent of the wealth, while the lowest 10 per cent account for 3 per cent of the wealth. Inequality in Shanghai is also marked – in 2019, the land values in the old, central areas of Jingan, Xuhai and Huangpu for residential flats were US\$8500 per m², whereas neighbouring districts that were once crowded with factories, such as Zhaibei, Yangpu and Putuo, were described as hubs for 'poor wretches'.

The Master Plan hopes to cap Shanghai's population at 25 million. This will be achieved by pushing excess people to marginal satellite towns. However, many of these have limited infrastructure and are under-developed with respect to schools and hospitals. Most of the employment remains in the city centre. Moving lower-income people to the edge of Shanghai leads to higher costs and more time for commuting, thereby reinforcing inequalities. The Master Plan also proposes the creation of more green spaces. For example, in 2023 the Yangshupu Power Plant became a public green space ([Figure 7.22](#)). This is a good example of sustainable development – using an old building premises for the greater good of the people.



▲ **Figure 7.22** In 2023 the Yangshupu Power Plant became a public green space

More than 50 per cent of people living and working in Shanghai do not have a Shanghai Hukou (registration) and thus they cannot draw on social support mechanisms. Many non-Shanghai residents are forced to live in satellite towns where social services are inferior, rather than in the city centre. In addition, Shanghai wants to attract foreign talent. The city

has a lower proportion of foreign skilled workers than London, New York and Los Angeles, but it wants to be seen as internationally competitive (Xie and Zhang, 2020).

Activities

- 1 Compare the challenges and opportunities brought about by urban growth in Shanghai.
 - 2 Assess the strengths and weaknesses of strategies to manage urban growth.
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Fieldwork

Urban fieldwork

The central business district (CBD) is the commercial and economic core of a city. It is the heart of the city and is the area that is most accessible by public transport. It has a number of characteristics:

- Multi-storey development: High land values force buildings to grow upwards, so the floor space of the CBD is much greater than the ground space.
- Concentration of retailing: Accessibility attracts shops with a high **range** and **threshold characteristics**. Department stores are located in CBDs; specialist shops occupy less accessible areas.
- Offices are concentrated in the CBD: Centrality favours office development.
- Vertical zoning is apparent: Shops occupy ground floors while offices occupy upper floors.
- Few people live in the CBD: A few luxury flats exist and there may be some artisans or university accommodation.
- Pedestrian flows and traffic restrictions are greatest in the CBD: Pedestrianisation has reduced areas for cars since the 1960s.

Possible hypotheses to test

Possible hypotheses that could be tested in the fieldwork include:

- The CBD is the most accessible area of the city for public transport.
- Building heights are highest in the CBD.
- Land use is mainly commercial – little land is used for residential or manufacturing purposes in the CBD.
- Pedestrian flows are greatest in the CBD.
- There are more parking restrictions in the CBD than elsewhere in a city.
- Vertical zoning is apparent.

Data collection

You could:

- compare different parts of the CBD, for example the core and frame, or compare the CBD with a suburban area, or take a transect from the centre of the CBD towards the edge of a small town/city
- use a base map of the city centre to mark off land use – make sure that you have a key to include different categories, such as commercial (high order), commercial (low order), finance (for example banks), restaurants, etc.
- record pedestrian flows at selected locations, counting the number of pedestrians passing you at 2-minute intervals
- analyse bus timetables to show the number and frequency of buses over several hours
- compare ground floor usage with that of upper floors to show vertical zoning.

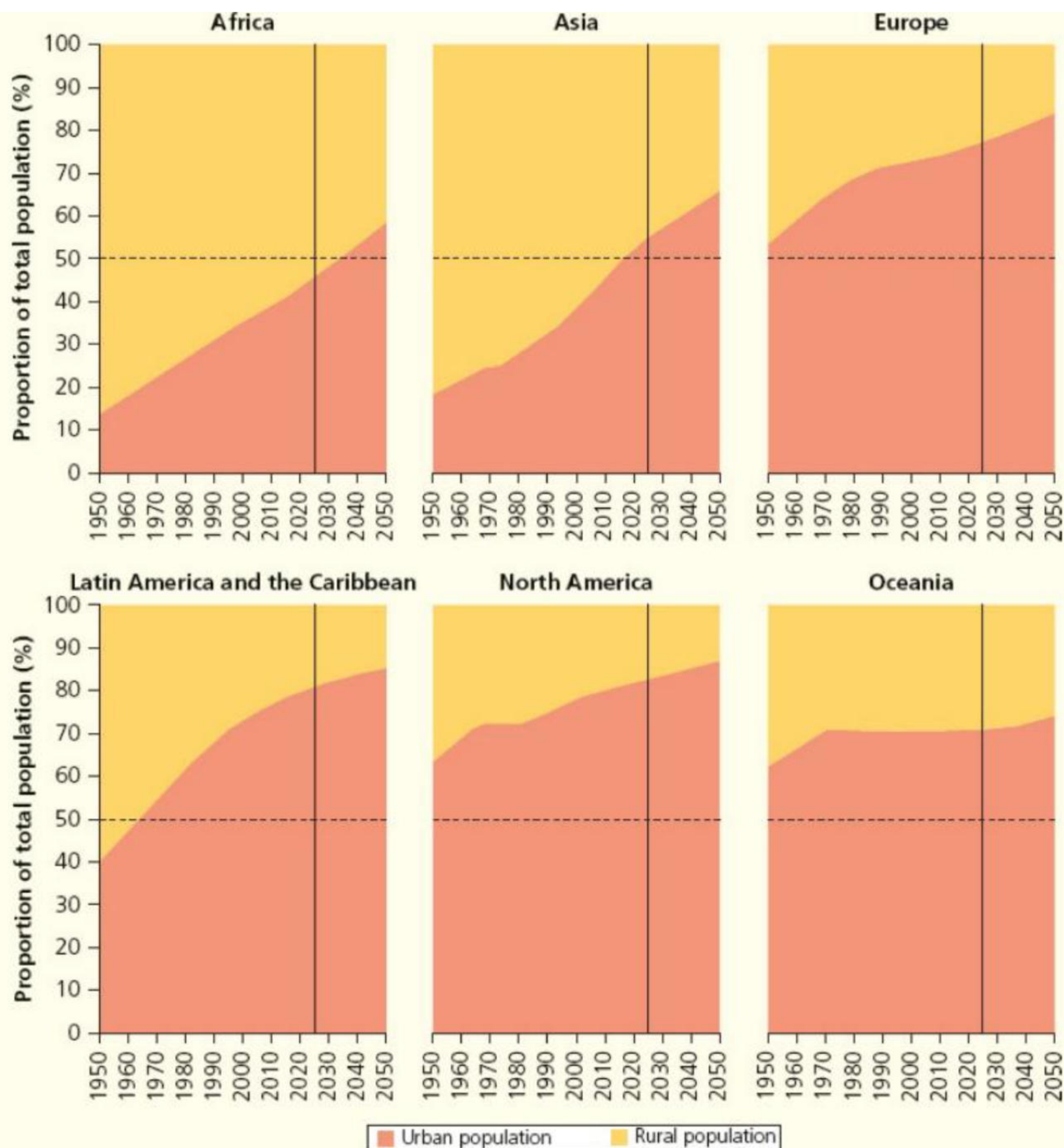
Along with these quantitative data forms you should take photographs on your phone to annotate later and provide qualitative data about the nature of the environment.

Presenting your findings

To present your findings you could use land-use maps, annotated photos, flowcharts of the number of buses, or located pie charts/bar charts to show pedestrian counts.

Practice questions

- 1 [Figure 7.23](#) shows changes in urban and rural populations by region between 1950 and 2050 (predicted from 2025).
 - a State the continent with the largest number of urban dwellers in 2025. [1]
 - b State the decade when 50 per cent of the population in Latin America and the Caribbean lived in urban areas. [1]
 - c Describe the trend in levels of urbanisation in Europe between 1950 and 2050 (predicted from 2025). [3]



▲ **Figure 7.23** Changes in urban and rural populations by region, 1950–2050 (projected)

2 [Figure 7.24](#) shows seasonal variations in air quality in Delhi. Explain the seasonal variations in air quality in Delhi. [5]



▲ **Figure 7.24** Seasonal variations in air quality in Delhi

3 [Figure 7.25](#) shows a development on the rural-to-urban fringe.

- Define the term 'rural-to-urban fringe'. [1]
- Suggest the advantages of the rural-to-urban fringe for new developments. [3]
- Assess the likely impacts of new developments in the rural-to-urban fringe on the environment. [4]

4 For a named example, assess the challenges and opportunities of urban living. [7]



▲ **Figure 7.25** A development on the rural-to-urban fringe